

第5次作业

4.2 1a) ~~u~~ $u = 10 \cos(\omega t - 10^\circ) \text{ V}$

4.3 由 $|Z| = \frac{U}{I} = \sqrt{R^2 + (\omega L)^2}$ 得

$$\frac{100\text{V}}{15\text{A}} = \sqrt{R^2 + (150\text{Hz} \cdot L)^2}$$

$$\frac{100\text{V}}{10\text{A}} = \sqrt{R^2 + (1100\text{Hz} \cdot L)^2}$$

$$R = 25.95\Omega \quad R = 5.09\Omega$$

$$L = 7.41 \times 10^{-5} \text{ H} \quad L = 0.086 \text{ mH}$$

4.5 $I_R = 1 \angle 0^\circ \text{ A}$

$$i_R = \sqrt{2} \cos(\omega t) \text{ A}$$

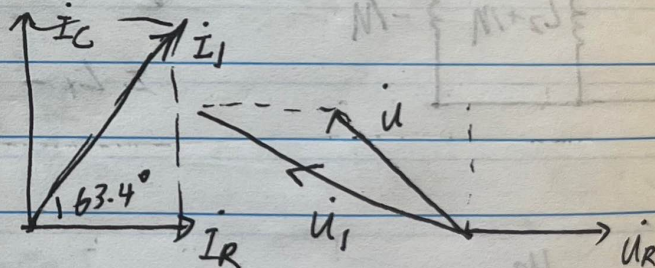
$$\dot{U}_R = \dot{I}_R \cdot R = 200 \angle 0^\circ \text{ V} = \dot{U}_C \quad u_R = 200\sqrt{2} \cos(\omega t) \text{ V}$$

$$\dot{I}_C = \frac{\dot{U}_R}{jX_C} = 2 \angle 90^\circ \text{ A} \quad \text{即 } i_C = 2\sqrt{2} \cos(\omega t + 90^\circ) \text{ A}$$

$$\dot{I}_1 = \dot{I}_C + \dot{I}_R = \sqrt{5} \angle 63.4^\circ \text{ A} \quad \text{即 } i_1 = \sqrt{10} \cos(\omega t + 63.4^\circ) \text{ A}$$

$$\dot{U}_1 = \dot{I}_1 \cdot j\omega L = 200\sqrt{5} \angle 153.4^\circ \text{ V} \quad u_1 = 200\sqrt{10} \cos(\omega t + 153.4^\circ) \text{ V}$$

$$\dot{U} = \dot{U}_1 + \dot{U}_R = 200\sqrt{2} \angle 135^\circ \text{ V} \quad \text{即 } u = 400 \cos(\omega t + 135^\circ) \text{ V}$$



4.7 由节点电压法

$$\left(\frac{1}{j\omega \cdot 1\text{F}} + \frac{1}{j\omega \cdot 1\text{H}} + \frac{1}{1\Omega} \right) \dot{U}_1 - \frac{1}{1\Omega} \dot{U}_2 = \frac{U_S}{j\omega \cdot 1\text{F}} - g\dot{U}_2$$

$$- \frac{1}{1\Omega} \dot{U}_1 + \left(\frac{1}{1\Omega} + \frac{1}{j\omega \cdot 1\text{F}} \right) \dot{U}_2 = I_S + g\dot{U}_2$$

$$\dot{U}_1 = 14 \angle 27^\circ \text{ V}$$

$$\dot{U}_2 = 3 \angle 5^\circ \text{ V}$$

$$\dot{U}_1 = 18 \angle 41^\circ \text{ V}$$

$$\text{即 } u_2 = 12 \cos(\omega t - 53^\circ) \text{ V}$$

$$\Rightarrow U_1 = (10j) V \quad U_2 = (10 - 10j) V \quad U_{12} = (-10 + 20j) V$$

$$\Rightarrow U_{12} = 10\sqrt{5} \cos(100t + 116.6^\circ) V$$

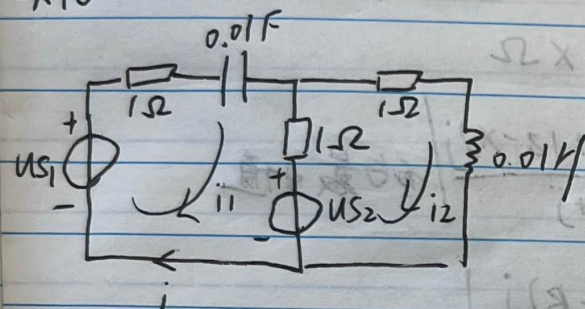
$$4.4 \text{ (b)} \text{ 设 } \dot{I}_C = 2 \angle 90^\circ A$$

$$\therefore \dot{U}_C = \dot{I}_C X_C = 60 \angle 0^\circ V \quad U = 60V$$

$$\therefore \dot{I}_R = \frac{\dot{U}_C}{R} = 1.2 \angle 0^\circ A$$

$$\Rightarrow I = \sqrt{I_C^2 + I_R^2} = 2.33 A$$

4.10



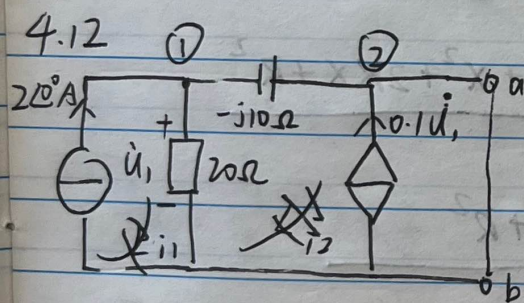
由回路电流法, 有

$$(1\Omega + 1\Omega + \frac{1}{j\omega \cdot 0.01F}) \dot{I}_1 + (-1\Omega) \dot{I}_2 = U_{S1} - U_{S2}$$

$$(-1\Omega) \dot{I}_1 + (1\Omega + 1\Omega + j\omega \cdot 0.01H) \dot{I}_2 = U_{S2}$$

$$\Rightarrow \dot{I}_2 = (5 - \frac{\sqrt{2}}{2}j) A \quad \dot{I}_1 = \frac{\sqrt{2}}{2} A$$

$$i = \cos(100t) A$$



开路时, 由回路电流

开路时, 由节点电压法

$$\text{有 } (\frac{1}{20\Omega} + \frac{1}{-j10\Omega}) \dot{U}_1 + (\frac{1}{j10\Omega}) \dot{U}_2 = 2 \angle 0^\circ A$$

$$(\frac{1}{j10\Omega}) \dot{U}_1 + (-\frac{1}{j10\Omega}) \dot{U}_2 = 0.1 \dot{U}_1$$

$$\Rightarrow \dot{U}_1 = -40V \quad \dot{U}_2 = (-40 + 40j) V$$

$$\text{开路电压 } \dot{U}_S = (-40 + 40j) V$$

$$\text{短路时有 } (\frac{1}{20\Omega} + \frac{1}{-j10\Omega}) \dot{U}_1 = 2 \angle 0^\circ A$$

$$\Rightarrow \dot{U}_1 = (8 - 16j) V$$

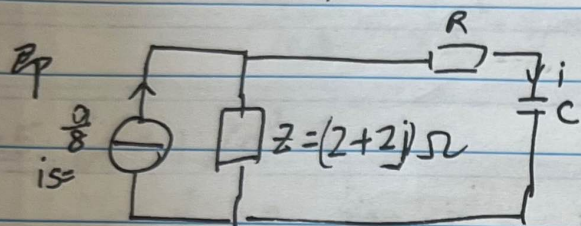
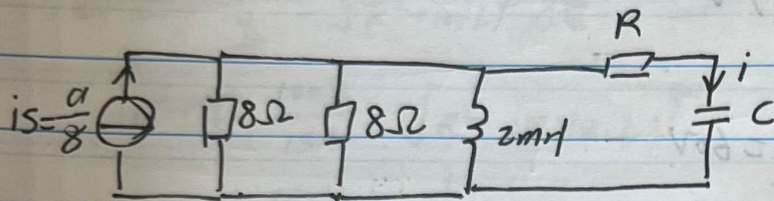
$$\therefore \dot{I}_S = \frac{\dot{U}_1}{-j10\Omega} + 0.1 \dot{U}_1 = (\frac{12}{5} - \frac{4}{5}j) A$$

$$\therefore \dot{Z}_S = \frac{\dot{U}_S}{\dot{I}_S} = (-20 + 10j) \Omega = 10\sqrt{5} \angle 153.4^\circ \Omega$$

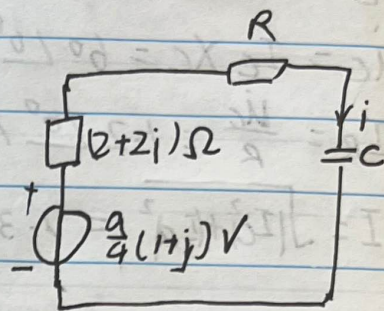
$$U_S = 40\sqrt{2} \angle 135^\circ V$$

4.14 设 $u_s = \sqrt{2} V$

电压源 $\rightarrow$ 电流源有



电流源 $\rightarrow$ 电压源 即



设 $C = -jX \Omega$ 设 $C = -jX \Omega$

$$\Rightarrow i = \frac{\frac{\sqrt{2}}{4}(1+j)}{(2+R) + (2-X)j}$$

即求 $\left| \frac{(2+R) + (2-X)j}{1+j} \right|$ 的最小值

$$|i| = \left| \frac{[(2+R) + (2-X)j](1-j)}{\sqrt{2}} \right| = \left| \frac{(4+R-X)^2 + (-X-R)^2}{\sqrt{2}} \right|$$

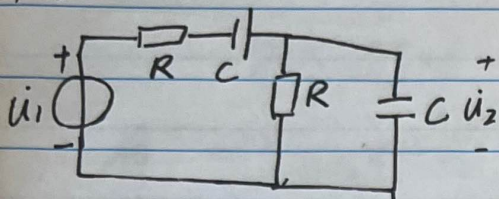
$$= \frac{(4+R)^2 - 2X(4+R) + X^2 + X^2 + 2RX + R^2}{2}$$

$$= \frac{2X^2 - 8X + 14 + R^2 + R^2}{2}$$

当 $X=2$ 时, 最小

即当 $\frac{1}{\omega C} = 2 \Omega$ 时 即 $C = 0.25 \text{ mF}$ 时, 最大

4.18



$$\begin{aligned}
 R // \frac{1}{j\omega C} &= \frac{\frac{R}{j\omega C}}{R + \frac{1}{j\omega C}} \\
 \therefore \dot{U}_2 &= \frac{\frac{R}{j\omega C}}{\frac{R}{j\omega C} + R + \frac{1}{j\omega C}} \dot{U}_1 \\
 &= \frac{\frac{R}{j\omega C}}{\frac{R}{j\omega C} + R^2 + \frac{2R}{j\omega C} - \left(\frac{1}{\omega C}\right)^2} \dot{U}_1 \\
 &= \frac{R}{3R + R^2(j\omega C) - \left(\frac{1}{\omega C}\right)^2 j\omega C} \dot{U}_1 \\
 &= \frac{R}{3R + j\omega C R^2 - \frac{j\omega C}{\omega^2 C^2}} \dot{U}_1
 \end{aligned}$$

同相时有 $\omega C R^2 - \frac{1}{\omega C} = 0 \Rightarrow \omega = \frac{R}{C} \Rightarrow \omega = \frac{1}{RC}$

即 $f = \frac{1}{2\pi RC}$

4.20 $W = UI \cos \varphi \Rightarrow \cos \varphi = 0.8 \quad \varphi = 37^\circ$

$|Z| = \frac{U}{I} = 3.6 \Omega \quad \therefore Z = 3.6 \angle 37^\circ \Omega$

$R = |Z| \cos \varphi = 2.88 \Omega \quad X_L = |Z| \sin \varphi = 2.16 \Omega = \omega L$

$\Rightarrow L = 6.87 \times 10^{-3} \text{ H}$

4.24 11) 设 $\dot{U} = 100 \angle 0^\circ \text{ V}$

$$\therefore \dot{I}_2 = \frac{\dot{U}}{R} = 5 \angle 0^\circ \text{ A}$$

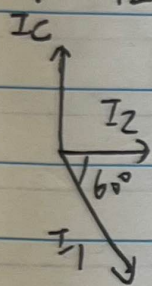
$$\dot{I}_1 = 10 \angle -60^\circ \text{ A}$$

$$\therefore \dot{I} = \dot{I}_1 + \dot{I}_2 = 5\sqrt{3} \angle -40.9^\circ \text{ A}$$

总功率因数 $\lambda = \cos \varphi = \cos 40.9^\circ = 0.76$

$$P = UI \cos \varphi = 1000 \text{ W}$$

12) 由相量图知



$$[(I_2 + I_1 \cos 60^\circ)^2 + (I_1 \sin 60^\circ - I_C)^2] = I^2$$

$$\Rightarrow I_C = 4.08 \text{ A}$$

$$\therefore I_C = \frac{U}{X_C} = U \cdot \omega C \Rightarrow C = \frac{I_C}{\omega \cdot U} = \frac{4.08}{314 \times 100} = 1.30 \times 10^{-4} \text{ F}$$

此时 $\cos \lambda = \cos \varphi = \frac{I_C + I_1 \cos 60^\circ}{I}$

$$\frac{4.08 + 10 \times 0.5}{5\sqrt{3}} = 0.76$$

4.27 由 $P = UI \cos \varphi = UI \lambda \Rightarrow I_1 = 5 \text{ A}$

设 $\dot{U}_1 = 200 \angle 0^\circ \text{ V}$

$$\therefore \dot{I}_1 = 5 \angle -36.87^\circ \text{ A}$$

$$\dot{I}_C = 2 \angle 90^\circ \text{ A}$$

$$\therefore \dot{I} = \dot{I}_C + \dot{I}_1 = \sqrt{17} \angle -14.04^\circ \text{ A} = 4.12 \angle -14.04^\circ \text{ A}$$

$$\therefore \dot{U}_R = \dot{I} R = 41.2 \angle -14.04^\circ \text{ V}$$

$$\therefore \dot{U} = \dot{U}_1 + \dot{U}_R = 240.2 \angle -2.39^\circ \text{ V}$$

$$\therefore U = 240.2 \text{ V} \quad I = 4.12 \text{ A}$$

$$4.29 \quad |Z_{AB}| = j\omega L + \frac{1}{j\omega C + \frac{1}{R_2}} = \frac{R_2 - R_2^2 \omega C j}{1 + R_2^2 \omega^2 C^2} = \frac{R_2}{1 + R_2^2 \omega^2 C^2} + j \left(\omega L - \frac{R_2^2 \omega C}{1 + R_2^2 \omega^2 C^2} \right)$$

最大功率时

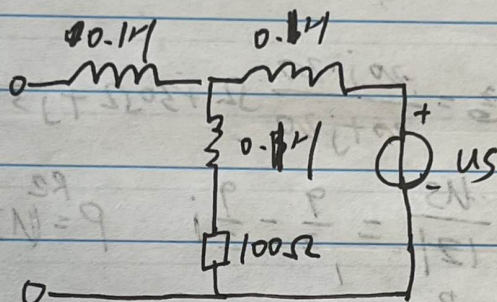
$$\text{有 } \frac{R_2}{1 + R_2^2 \omega^2 C^2} = R_S \quad \omega L - \frac{R_2^2 \omega C}{1 + R_2^2 \omega^2 C^2} = 0$$

$$\Rightarrow \cancel{C = 7.5 \times 10^{-11} \text{ F}} \quad \cancel{L = 1.875 \times 10^{-6} \text{ H}} \quad C = 1.936 \times 10^{-8} \text{ F}$$

$$P_m = \frac{U_S^2}{4R_S} = 2 \times 10^{-3} \text{ W}$$

$$L = 4.841 \times 10^{-4} \text{ H}$$

4.33 (b) 即等效电路

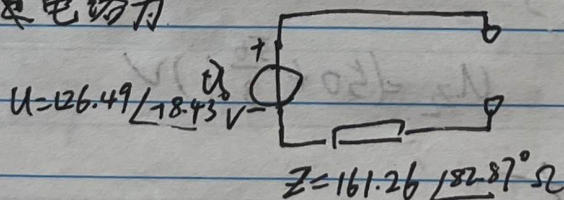


$$\text{开路电压 } U = \frac{100 + j100}{100 + j300} \cdot U_S = \frac{100 + j100}{100 + j300} \cdot U_S = 103.96 \angle -8.97^\circ \text{ V}$$

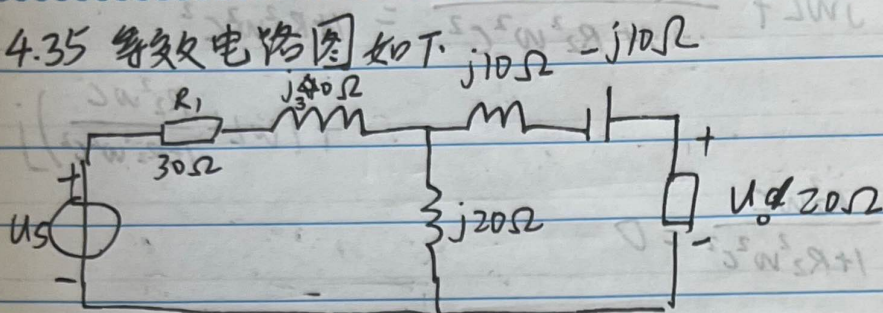
$$= 126.49 \angle -18.43^\circ \text{ V}$$

$$Z = \left(100j + \frac{(100 + j100) \cdot 100j}{(100 + j100) + 100j} \right) \Omega = 161.26 \angle 82.87^\circ \Omega$$

戴维南等效电路为



4.35 等效电路图如下



$$U_o = \frac{20 \cdot j20}{20 + j20} \Omega$$

$$\frac{20j20}{20+j20} \Omega + 30\Omega + j30\Omega$$

$$U_s = 90 \angle 0^\circ \text{ V} \quad \therefore \text{有效值为 } 90 \text{ V}$$

$$P_{\Sigma} = \frac{U_s^2}{|Z_{\Sigma}|}$$

$$Z_{\Sigma} = \frac{20j \cdot 20}{20+j20} \Omega + 30\Omega + j30\Omega = (40+40j)\Omega$$

$$\therefore P_{\Sigma} = 229.1 \text{ W} \quad I = \frac{U_s}{|Z|} = \frac{9}{2} - \frac{9}{2}j \quad P = \text{Re}(UI) = \frac{1360}{1620} \text{ W}$$

$$P_R = \frac{U_o^2}{R} = 405 \text{ W} \quad |Z_L'| =$$

$$\frac{P_R}{P_{\Sigma}} \approx 17.7\% \quad 25\%$$

4.37 等效阻抗为 $n^2 \cdot |Z_L|$

$$P_{\text{最大}} \text{ 时有 } |Z_L'| = R_S \Rightarrow n = \frac{1}{2}$$

$$Z_{R_{\Sigma}} = R_S + Z_L = 5\Omega + 14 + 3j \Omega = (19+3j)\Omega$$

$$\therefore I = \frac{U}{Z_{R_{\Sigma}}} = (10 - \frac{10}{3}j) \text{ A}$$

$$U_{Z_L} = (50 + \frac{50}{3}j) \text{ V}$$

$$\therefore P = I^2 R_{\Sigma} = \frac{5000}{9} = 555.6 \text{ W}$$

$$P = \text{Re}(UI) = \frac{4000}{9} \text{ W}$$